

Analysis Of TRANSFORMER INRUSH CURRENT Using Java

ASHISH BHATIA

Power transformers are essential part of a power system network.

A transformer is used at the transmission side to step up the voltage level and at the distribution side to step down the voltage. A thorough knowledge of the transformer's performance under various operating conditions is fundamental in determining the power system reliability.

One of the major issues in maintaining the power-supply quality is the disturbance caused by transformer energisation. Inrush current is the magnetising current that occurs during switching of an unloaded or lightly loaded transformer onto the electric grid. Due to non-linear magnetic characteristic of the transformer core, transformer energisation results in inrush currents of a

high magnitude, which, under certain conditions, may reach several times the rated current of the transformer.

Inrush current is highly non-sinusoidal in nature and thus rich in harmonics. It may damage the transformer itself, affect relay operation and reduce power quality due to the presence of harmonics. Inrush current is a problem as it interferes with the circuit operation; in a digital world, there is zero tolerance for power interruptions.

So accurate determination of inrush current is desirable to predict potential problems when an unloaded or lightly loaded transformer is switched onto the power supply. This is essential to devise correct methods to curb inrush current.

Factors affecting the magnetising inrush current

The transient electromagnetic state causing inrush current in a transformer when it is connected to the power supply, depends on the following factors:

1. Voltage angle when the transformer is switched on
2. Magnitude and polarity of residual flux in the transformer core at the instant of energisation, that is, residual flux density (B_r)
3. Total resistance of the primary winding circuit including system resistance
4. Inductance of the air-core in between the energised winding of the transformer

5. Geometry of the transformer core, that is, axial length of the energised winding in the transformer core, that is, axial length of the energised winding (H_w); area enclosed by mean turn of the winding, that is, mean diameter of the energised wind-

ing (D_m); and number of turns of the excited winding

6. Maximum flux-carrying capability of the core (B_m)

7. Saturation flux density of the core (B_s)

8. Capacity of the transformer and its primary/excited voltage value

A typical hysteresis curve showing relationship between magnetic field strength (H) and flux density (B) for the transformer core material is shown in Fig. 1.

Residual flux density (B_r). In Fig. 1, the hysteresis loop clearly shows residual flux density. When a transformer is de-energised, some level of flux remains in the transformer core. The actual value of flux is based on the alignment of the steel's magnetic moments, and can be found from magnetic hysteresis loop of the transformer core. Residual flux may therefore be positive or negative in value, and is typically 30-80 per cent of the core's maximum flux. When the transformer is energised, this residual flux is added to the flux driven by the exciting voltage.

Saturation flux density (B_s). The hysteresis loop shows when a high magnetising force is encountered, a point is reached where further increase in H does not cause an increase in B . This point is known as the saturation point of that material. Saturation flux density B_s and magnetising force required to saturate the core (H_s) are shown through dashed lines in Fig. 1.

Peak flux density (B_m). This is the peak value of designed steady-state flux density in the core.

The inrush current phenomenon including the effect of residual flux is shown in Fig. 2. A typical inrush current waveform is shown in Fig. 3.

Inrush current model

In this software, the technique suggested by T.R. Specht, popularly known

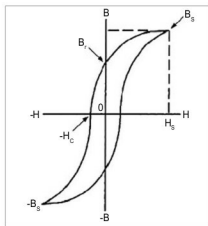


Fig. 1: Hysteresis

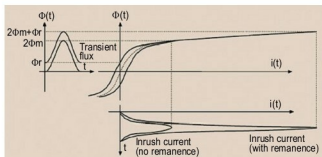


Fig. 2: Inrush current phenomenon